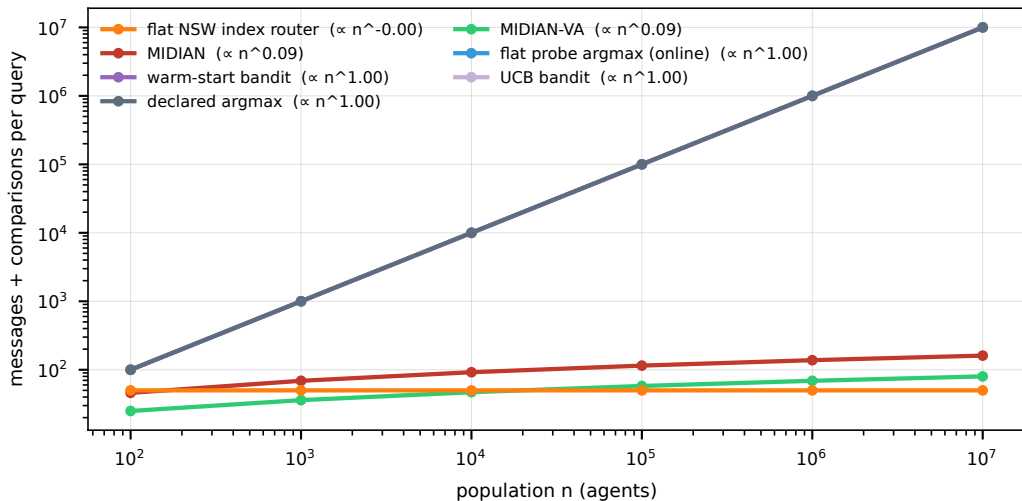


I efficiency: MIDIAN routes in $O(\log n)$ work and, after a linear one-off build, $\sim$ no energy per query (* energy = scripts/energy.py estimate)

(a) routing work per query: MIDIAN $\sim \log n$, flat methods $\sim n$ (bernoulli, $b = 3$, exact ledger)



(b) energy per query (*): one probing build, then ~ 0.01 J / query vs an LLM call per query

